A Survey of Using Generative Models to Perform Reverse Engineering

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Abstract

This research examines how generative models, GANs, VAEs and Transformers and Diffusion Models can be used to automate and improve reverse engineering in the software, hardware and protocol realms. The research includes the analysis of the current sources and compares the state-of-the-art strategies to describe the contribution of the models to processes like binary decompiling, circuit reconstruction, and protocol inference. Although generative AI has a great effect of increasing speed and scalability, there are also concerns related to accuracy, model hallucination, adversarial risks, and dual-use misuse. Each model has some strengths and weaknesses, which are stated in the study, and there is an increasing tendency in hybrid approaches to include symbolic reasoning and generative procedures, which enhance reliability and interpretability. Issues of ethical integrity, such as intellectual property and governance.

Keywords: *Generative Models, Reverse Engineering, Transformers, GANs, VAEs, Diffusion Models, Software Decompilation, Hardware Reconstruction, Protocol Inference, Large Language Models, Model Hallucination, Explainable AI, Adversarial Risks, Intellectual Property*

1. Introduction

Reverse engineering (RE) is the practice of taking a system and breaking it down to deduce its structure, functionality and design, often without needing to consult the source code or documentation. In the traditional sense, reverse engineering demanded experts' knowledge, manual inspection and a mix of debugging, disassembly and schematic tracing methodologies [1]. RE has many usages in the field of software engineering, such as the analysis of legacy software or malware, or of proprietary binaries. In hardware, it is the rebuilding of an integrated circuit (IC) or printed circuit board (PCB) or microarchitecture. On protocols, RE makes sense of uncharted communication protocols to test interoperability or assess vulnerability. This area has also changed over the years to be more automated than manual over time, but the issues of incomplete recordings, obfuscation and anti-analysis continue to affect it. Due to the emergence of Artificial Intelligence (AI) and specifically generative models, an additional sphere of automation has arisen. Generative models are built to generate new data that can be similar to a training distribution. This includes:

Generative Adversarial Networks (GANs) that have high-fidelity output and train two neural networks to

compete with each other. ***Variational Autoencoders*** (VAEs) are a way of modelling data distributions and can reconstruct inputs by latent encoding [2]. ***Large language models*** (LLMs) and other transformers are particularly good at working with sequences, like code, binary instruction sets, and network protocol logs. ***Diffusion models*** that are iterative and produce outputs through successive refinement of noise have become popular in tasks where fine-grained control is necessary. These models were recently studied in the reverse engineering setting, such as automatic decompilation, source code generation, circuit layout prediction, and protocol reconstruction. This ability to train generalisations on big-scale data puts them particularly in a situation to estimate non-documented or closed actions of systems, which offers risks besides the prospects in the worlds of cybersecurity, intellectual property, and ethical governance.

Research Aim and Objectives

Aim

The research aims to explore the role and effectiveness of generative models in enhancing and automating reverse engineering processes.

Objectives

- To analyse different generative models applied in reverse engineering of software, hardware and cybersecurity.
- To examine how generative models help with accuracy, speed and automation in several reverse engineering cases.
- To examine ethical, security and intellectual property issues related to generative AI in the process of reverse engineering.

Scope of the study

The study is based on the domain of computer science, cybersecurity, and artificial intelligence and, more specifically, is related to the applicability of generative models in reverse engineering (RE). As systems grow in complexity and grow more obfuscated, the conventional reverse engineering practices, as effective as they may be, face issues of scalability, interpretability and efficiency [3]. The emergence of generative AI presents the unprecedented opportunity to support or automate the essential RE processes. Computer science Generative models are revolutionising the way software is analysed, decompiled and protocol reverse engineered. This includes the instances of source code inversion of binaries or application programmer interfaces out of interfaces that are not documented through transformers. The ability to consume and reason across sequential data has rendered language models quite useful in understanding and generating behaviour within frameworks.

Generative AI has the potential to substantially boost defensive operations, including malware examination, vulnerability observation, and checking of legacy systems, on the one hand. Reproduce proprietary systems using the same models, get around protections or even create adversarial code, which brings ethical and legal implications. Relevance to legacy software recovery is also stressed in the study, where original documentation or source code is not available. Generative models may be used to fill gaps by approximating the original design to allow modernisation or integration. These models can display concealed functions, control flows, or signatures that are avoided by

standard malware identification algorithms in malware analysis.

2. Related Work

2.1 Software Reverse Engineering

Software reverse engineering (SRE) involves analysing a compiled program or obfuscated code to reconstruct high-level forms of the representation, such as the source code, control flows, or architectural understanding. Due to the emergence of Large Language Models (LLMs) and transformer-based architectures, major steps have been taken towards the automation of tasks that were previously performed by experts in a manually supervised way [4]. On very large code corpora, LLMs are already trained to accomplish binary decompilation, function name retrieval, and malware behaviour analysis. Such models have been promising in restoring pseudo-code or even source approximations of the original in stripped binaries. This is especially handy in the case of malware analysts or forensic engineers who encounter obfuscated executables.

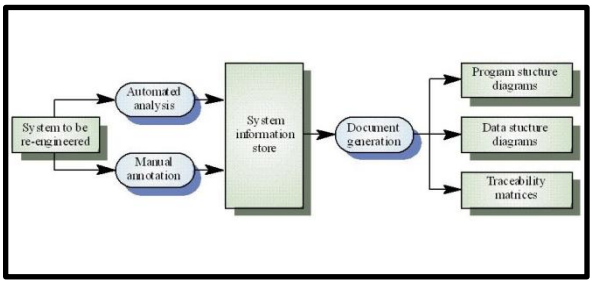


Fig 1: Reverse Engineering process

New tools and databases give aid to these remarkably. A deBinVul is a dataset that is proposed to test the performance of the model on decompiled binaries and can test the degree to which LLMs recognise the vulnerability based on the machine code [5]. Some other framework combines the transformer models with symbolic execution to manage the control flow recovery and type inference. There are limitations of the LLM-based tools despite their promise. They are very slow when applied to non-C/C++ binaries, and suffer severe performance regressions with every major revision. LLMs are also prone to hallucinating, which is problematic in critical tasks such as malware analysis or vulnerability patching. In addition, explainability and trustworthiness are still the issues. It requires analysts to know how a model gets to the prediction, particularly when they interpret malicious code sections. Although other studies aim at making interpretations by using attention mechanisms or prompt tuning, the black-box behaviour of many LLMs is impeding progress. With the development of the sphere, stronger benchmarking, safety restrictions, and the hybrid approaches are bound to become paramount to the future of software reverse engineering.

2.2 Hardware Reverse Engineering

Hardware reverse engineering (HRE) is the study of integrated circuits (ICs) and printed circuit boards (PCBs), and logic designs to recreate their design and their operation. Conventionally done by hand observation, delayering and microscopic image, HRE has been increasingly complicated by increased sophistication and security of hardware [6]. However, generative models have recently started to help automate this. The models can be used to reconstruct the netlist, layout and clone an IC based on the learning of an existing hardware dataset and generate a schematic(or gate-level) design that is an approximation of the original architecture.

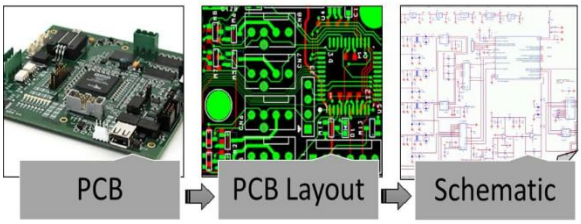


Fig 2: Hardware Reverse Engineering

There are notable publications that form the academic background in this respect. The algorithms used in IC reverse engineering, such as logic extraction, deobfuscation or hardware cloning techniques [7]. More recent entries, Model Reverse Engineering Threats in the direction of side-channel and fault-injection attacks, showing how models can be used to train and infer sensitive hardware configurations or internal parameters based on observable data. These and other works highlight the effectiveness with which generative AI is facilitating more scalable, though potentially hazardous, approaches to reverse engineering at a hardware level.

A recent trend is the introduction of HDL-generating transformers, in which transformer-based models are applied to automatically generate Verilog or VHDL code given more abstract specifications or obfuscated logic [8]. These systems have proven qualities of being quite capable of rebuilding logic and circuit operation with minimal human interaction. Hardware training data is more limited than software training data, and many ICs incorporate proprietary designs which are not readily available to train models. Additionally, it is resource-consuming to physically verify created hardware. However, generative models will play an essential role in decreasing and increasing the pace of hardware reverse engineering.

2.3 Protocol and API Reverse Engineering

Reverse engineering of protocols and APIs is directed at the work of recreating undocumented communication protocols, and system application program interfaces, essential in software interoperability, malware analysis and system integration of legacy systems. This traditionally has been a labour-intensive process involving a lot of manual monitoring of input/output (I/O) behaviour and traffic analysis [9]. With the emergence of transformer-based models, and especially those models trained on sequence prediction and pattern recognition, however, there is a shift towards automated protocol inference.

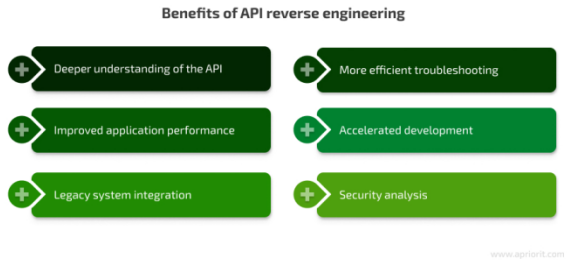


Fig 3: Benifits of API reverse engineerinng

Recently, there has been a demonstration of how generative models may be used to recreate protocol state machines, that is, how a system moves between various states as it communicates. Treating I/O sequences as natural language-like data leads transformers to find useful in determining repetitive structures, shifts, and oddities in protocol exchanges [10]. This is of special interest when applied to proprietary or malicious protocols, which either lack documentation or are deliberately obscured. Although the automation of the process increases its scalability, it is still difficult to differentiate between a legitimate protocol behaviour and noise and to capture edge-case scenarios. However, generative

models are increasingly becoming a critical reverse engineering tool of APIs and protocols, particularly with the support of symbolic and statistical analysis tools to improve accuracy and interpretability.

2.4 Ethical, Legal, and Security Considerations

Although the use of generative AI in reverse engineering improves this process substantially, it also presents ethical, legal, and security issues. The dual-use nature of these technologies is one of the most critical issues. Most models used for automating decompilation or re-synthesising circuits could be applied both to legitimate research and to malicious such as stealing intellectual properties or spying. Proprietary code, hardware logic, or communication protocols: The capacities of generative models to reproduce proprietary code, hardware logic, or communication protocols blur the differences between innovation and infringement [11].

Intellectual property protection is another big problem. The model can replicate content under copyright or other proprietary data when trained on the latter. This brings legal concerns in terms of ownership and liability regarding the AI-generated outcomes and particularly where designs replicate or are similar to the designs. Technically, generative models, in turn, are also subject to security risks. Diffusion models and transformers possess the potentially devastating flaws of adversarial attacks, data poisoning, and model inversion, which might tamper with their outputs or reveal their training data [12]. The growing research and development in the form of machine unlearning and better explainability solutions are thus expected to address these issues by improving the transparency and accountability of models. Nevertheless, the prudent use, government regulation, and ethical considerations are still significant in addressing the dangers of AI-based reverse engineering.

3. Comparison, Findings and Insights

3.1 Comparative Analysis of Generative Models

Generative models have become especially useful in many areas of reverse engineering, automating, making more precise, and speeding up processes formerly time-consuming. All architecture models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers and Large Language Models (LLMs), Diffusion Models have unique advantages and are more appropriate in different reverse engineering tasks. Generative Adversarial Networks (GANs), since they are especially suitable for visual reconstruction tasks, are quite potent in hardware reverse engineering [13]. GANs are a framework of competing generator and discriminator, and GANs allow the generator to generate a highly realistic model of learned data distributions. Examples of GAN usage include generating IC layouts or microscopic image-based reconstructions of partial or noisy input in the field of circuit analysis, creating a style transfer in which GANs generate reconstructions of images in the style of the input images to form a style transfer, and generating outputs in a different style or modality in the field of image-to-image translation.

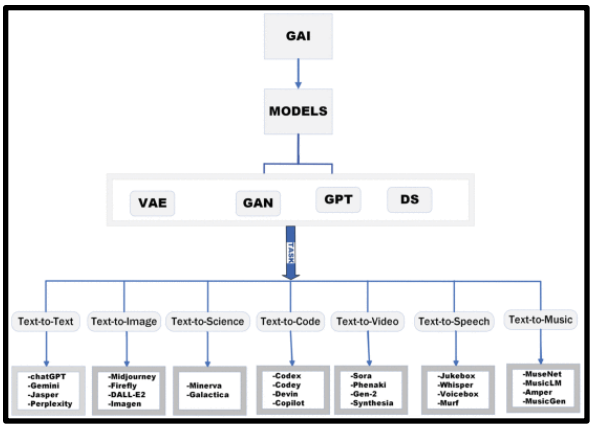


Fig 4: GAI models comparison

Figure 4 sows categories of generative models that is helpful in data compression, dimensional reduction and learning of latent representation is the Variational Autoencoders (VAE). Proto-types VAEs may simplify succinct representations of systems behaviour or physical structure to model protocols or low-level hardware [14]. Their probabilistic model allows variational auto-encoders to produce a wide range of outputs, so they are appropriate when examining the design variants or reconstruction of uncertain system configurations. Nevertheless, VAEs generally generate blurrier or less accurate results than GANs, and their reconstruction quality may not be high enough to allow careful software reverse engineering or even precise hardware emulation.

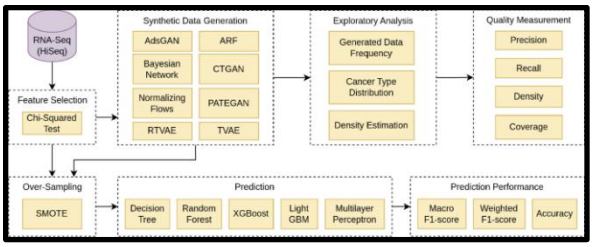


Fig 5: Comparative Analysis of Tabular Generative Models

Large Language Models (LLMs) and Transformers have changed software reverse engineering. They model long-range dependencies in code, binary sequences, or protocol exchanges through their self-attention-based architecture. These models perform well on binary decompiling and source code recovery as well as API inference. Function names, variable types, or control flows can be predicted with the aid of LLMs such as GPT variants, as fine-tuned LLMs on decompiled binaries [15]. They are very effective in reverse engineering complex malware or undocumented communication protocols since they can model sequential logic so well. Besides, LLMs facilitate the use of natural language communication, and analysts can now query models with sentences in the human form.

A relatively newer invention, Diffusion Models is attracting some attention due to their Posterior stability, controllability, and model output quality. Such models are generative and, through an iterative process, denoising of random noise will lead to structured output, allowing great control over the generative process. Diffusion models are under study in reverse engineering to recover structured hardware design, to generate visual artefacts, and even to decode obfuscated logic[16]. They are computationally expensive to train, and even when trained, their results are not as unstable or of as poor quality as GANs. Moreover, the ecosystem surrounding the use of their software RE is still nascent, as compared to the LLMs.

3.2 Strengths and Limitations in Applications

The use of generative models in reverse engineering offers radical advantages in speed, accuracy, and automation, but also significant drawbacks to their

robustness, hallucination, and generalisation. These forms of trade-offs are crucial to know when implementing such models for reverse engineering work in software, hardware and protocols. Among the most apparent advantages of generative models, they can distinguish the automation of repetitive and time-consuming tasks, which completely speed up reverse engineering workflows. As an example, transformer-based models can quickly produce readable pseudo-code of binary input, saving hours of manual instruction trace that analysts would otherwise spend [17]. GANs and diffusion models also allow reconstruction of high-fidelity circuit layouts or obfuscated images given a little visual input, used by hardware analysts to restore complex chip designs.

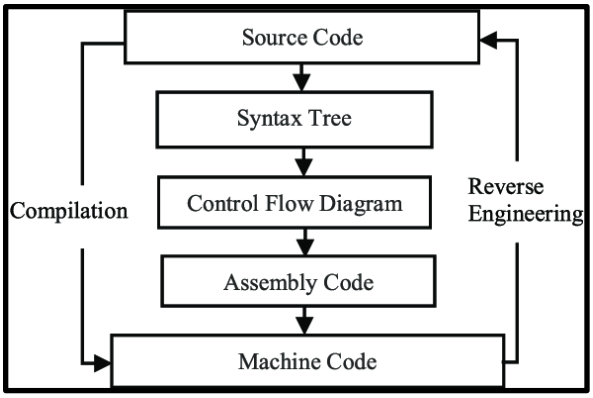


Fig 6: Flow chart of reverse engineering

Regarding accuracy, such models serve well when they are trained with data sets that are detailed and specific to the domain. As an example, language models that are fine-tuned on security-specific code corpora can make predictions of vulnerabilities or predict function names with great precision. But, such precision can crash down quickly when the data on which it is trained does not match the actual distribution. Most generative models have a problem with generalising to previously unseen systems, especially when trying to analyse binaries in newer programming languages, custom obfuscated hardware or new protocols that have distinct state evolutions [10].

One of the major drawbacks has to do with the problem of hallucination, wherein a model produces output that is realistic yet erroneous. In the case of reverse engineering, such mistakes are not just a nuisance; they may misguide the analysts, provide false clues, and may provide a security delusion [16]. It is more so when it comes to malware analysis or vulnerability detection, as in those cases, false positives or negatives might be disastrous to operations. This issue is especially likely to affect transformers and LLMs, unless they are strictly validated. Although diffusion models do not have a problem of unstable output, they are computationally demanding, and they have yet to be fully applied in reverse engineering. Despite their capacity to produce visually pleasing results, GANs are associated with training instability and an inability to interpret.

3.3 Trust, Interpretability and Adversarial Risks

Generative models are still grappling with increasingly critical issues of trust, interpretability, and vulnerability to adversarial manipulation that is particularly important in reverse engineering. The adoption of these models in recovering sensitive information about the system is rising, which is a great security threat to the model with variations like model inversion, Extraction attacks, and data leakage [18]. Adversarial vulnerabilities are computed from maliciously made inputs that will trick a model into giving inaccurate or misleading outputs. Within the context of reverse engineering, this may refer to

passing tampered with binaries or circuit traces to a model to produce a forged, degraded reconstruction or evade sensors. Model inversion attacks go even further to allow malicious actors to reverse-engineer the model itself, and may be able to extract sensitive information used in the training process, which could include proprietary code or even hardware blueprints.

The other big issue is the interpretability of generative outputs. Numerous well-performing models are black boxes, in particular, transformers and GANs and provide minimal insight into the mechanism of making predictions. This poses a trade-off between interpretability and performance: more explainable models can produce worse performance, and more accurate models can be less scrutable or testable [14]. This opacity can undermine user confidence in a high-stakes field such as reverse engineering. There are attempts to advance explainable AI (XAI) methods specific to generative systems, such as attention visualisation, gradient-based attribution, and example-based explanations. However, a balance between trust, security and performance is a worthwhile challenge. During this process, sound auditing systems and adversarial robustness will be important factors in assuring the reliability and safety of the tools in reverse engineering pipelines, as generative AI becomes more deeply integrated.

3.4 Cross-Domain Insights and Trends

Generative models have been applied in reverse engineering across a wide variety of areas, including software decompilation, hardware reverse engineering and reverse engineering of network protocols. But throughout all these examples, some general areas of gaps and challenges appear, specifically where data is concerned, model validation, and integration of tools. Dataset availability and diversity are one of the main limitations. Existing datasets are often limited to a specific subset of C/C++ binaries, common hardware configurations, or completely documented protocols, making coverage of contemporary programming languages (Rust, Go) or complicated IC layouts more difficult, or encrypted/custom protocols [16]. This limits the generalisability of models and their applicability to real-life situations. In addition, the reproducibility of results is also a problem, with many of the models being based on proprietary data sets or lacking publicly benchmarked models, necessitating the need to check performance or compare methods between papers.

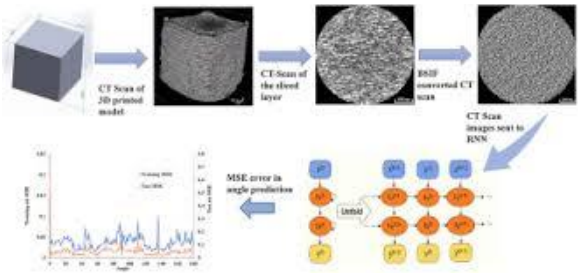


Fig 7: Reverse engineering of additive manufactured composite

The other theme is the increasing desire to have the hybrid reverse engineering methodologies that integrate generative learning and symbolic reasoning. The symbolic execution, taint analysis and SMT solvers are traditional tools: they are precise and traceable, whereas generative models are fast and they generalise patterns. The complementarity of these methods, a combination in which generative models propose hypotheses that are checked by symbolic engines, has proven to be effective on both efficiency and reliable grounds. One can also observe

a tendency to create modular RE frameworks, where models can be specialised to subtasks and be coordinated in a wider system.

4. Future Work

With generative models in the process of transforming and being involved in a reverse engineering process, it becomes essential to develop research in specific niches in order to remove the existing restrictions and increase the actual and ethical use in applications. This section describes four primary subject areas of future work to increase accuracy and reliability, make interpretability more effective, set up governance plans, and explore new technological frontiers.

4.1 Addressing Hallucination and Accuracy Problems

One of the most urgent problems of the generative models in reverse engineering is the problem of hallucination. The errors can lead to misinforming engineers, falsely detecting vulnerabilities, or generating invalid reconstructions in the case of software or hardware reverse engineering, and this can be dangerous in such a high-stakes setting as malware analysis or security review. The future research needs to concentrate on uncertainty-based approaches, which can quantify the certainty of the models [15]. This can accommodate layers of probabilistic modelling, Bayesian neural networks or even the introduction of uncertainty quantification mechanisms into the pre-existing generative structures. When it comes to human reviewers, getting not only outputs but also a feeling of where the confidence in the results is, the human reviewer can make a better decision on the credibility of the results.

Also, there is a need to pay more attention to timely engineering solutions and follow-up designs. The output can be directed to more accurate and relevant results by polishing the input prompts and entrenching constraints. The safety net approach of post-processing with static analysis, symbolic execution or logic validation can be used to fix or check model results before operational use [16]. The most reliable way forward might be a combination, in which generative models are utilised as a source of creativity and deterministic tools are used to provide certainty.

4.2 Enhancing Interpretability and Explainability

Most generative models are opaque, making it a major obstacle to trust and use, especially in applications such as reverse engineering, where traceability and accountability are non-negotiable. In future research, one should aim to embed explainability mechanisms within the architectures of the model or the deployment pipeline. A potentially fruitful avenue is a combination of symbolic reasoning and generative models, in which intermediate steps in the logic can be revealed and verified.

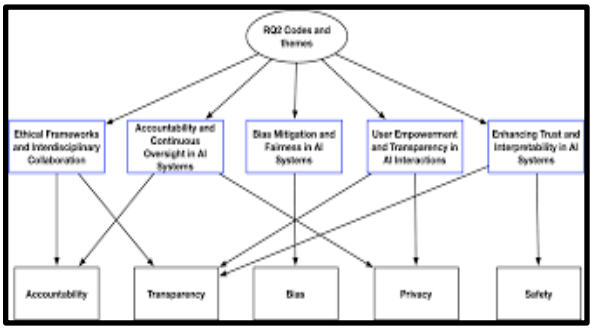


Fig 8: Ethical Concerns of Generative AI and Mitigation Strategies

This mixed method has the capacity of giving a better insight into how a model ended up at a given reconstruction or prediction. Explainable AI not only contributes to the confidence that users gain in model outputs but also contributes to debugging model failures, decreasing the risk of hallucinations, and enhancing the training process [14]. Increased interpretability will eventually assist in safer and more efficient application of generative models in RE tasks.

4.3 Ethical Governance and Regulation

As generative AI systems will be able to take the form of proprietary software, hardware design, or secret protocols, the danger of abuse of a dual-use increases drastically. In the future, the development of ethical frameworks of governance to address and control such risks should be included. This includes the formal process of risk evaluation of dual-use technologies, especially within the defence, medical devices, and critical infrastructure. Model watermarking and auditing routines must also be done with great strength. Watermarking enables the determination of whether an artefact created relates to a particular model, a factor which can be leveraged in terms of ownership and responsibility [15]. Model usage logs, query patterns, and access trails can be tracked by the auditing mechanism, so the approach is transparent and allows for investigating misuse.

5. Conclusion

This paper has reviewed how the use of generative models, GANs, VAEs, Transformers, and Diffusion Models has contributed to advancing the art of reverse engineering in software, hardware, and protocol space. Such models have the important benefit of being fast, scale-friendly and automatic, making them useful in tasks like binary decompilation, circuit synthesis and protocol inference. Nevertheless, their application also imposes serious drawbacks, such as hallucination, interpretability, adversarial vulnerability, dual-use ethical ambiguity, and intellectual property.

Comparative analysis demonstrates that GANs and diffusion models are more appropriate to use with the visual and hardware reconstruction, whereas transformers and LLMs are more suitable in terms of software and protocol analysis. VAEs provide a trade-off between structure and generalisation with the restriction of output fidelity. Cross-domain tendencies show a tendency to turn to hybrid systems that integrate symbolic logic with generative learning as an approach to enhancing reliability and control. To achieve the potential of generative models in reverse engineering, the research should aim at enhancing model trustworthiness, acquiring better explainability, solving legal risks, and expanding applicability to new domains of technologies. These models have the potential to provide engineers, security analysts and researchers with new and potent possibilities when used responsibly.

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